A Study of VideoAgent Performance

Qwen3-VL-2B-Instruct | *Timestamp search video reasoning and event onset*

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Abstract

This study evaluates timestamp-based content search, video reasoning, and detection of the first occurrence of an event in VideoAgent v6. It presents a secondary comparative analysis of results from two simple60 audits. The main comparison covers adaptive and corrected uniform execution: the same 60 questions in each condition, yielding 120 results. Another 60 results from the original uniform implementation serve as an engineering baseline before the fixes. In total, the study examines 180 results for 60 unique questions about one video.

Strict semantic success is 19/60 for adaptive and 21/60 for corrected uniform. However, adaptive more often gives a correct core answer on answerable questions: 25/45 versus 16/45. Neither mode produces a fully correct event-onset answer. Corrected uniform requires 22.8% more total question time. The original uniform implementation records 53 out-of-memory (OOM) failures and two frame-contract failures; these are distinguished from failures to understand the video.

Measure	Adaptive 60	Uniform fixed 60
Fully correct answers and correct abstentions	19/60 · 31.7%	21/60 · 35.0%
Partially correct answers	19/60 · 31.7%	6/60 · 10.0%
Correct core on answerable questions	25/45 · 55.6%	16/45 · 35.6%
Fully correct event-onset answers	0/15	0/15
Technical failures of the final response	8/60 · 13.3%	7/60 · 11.7%

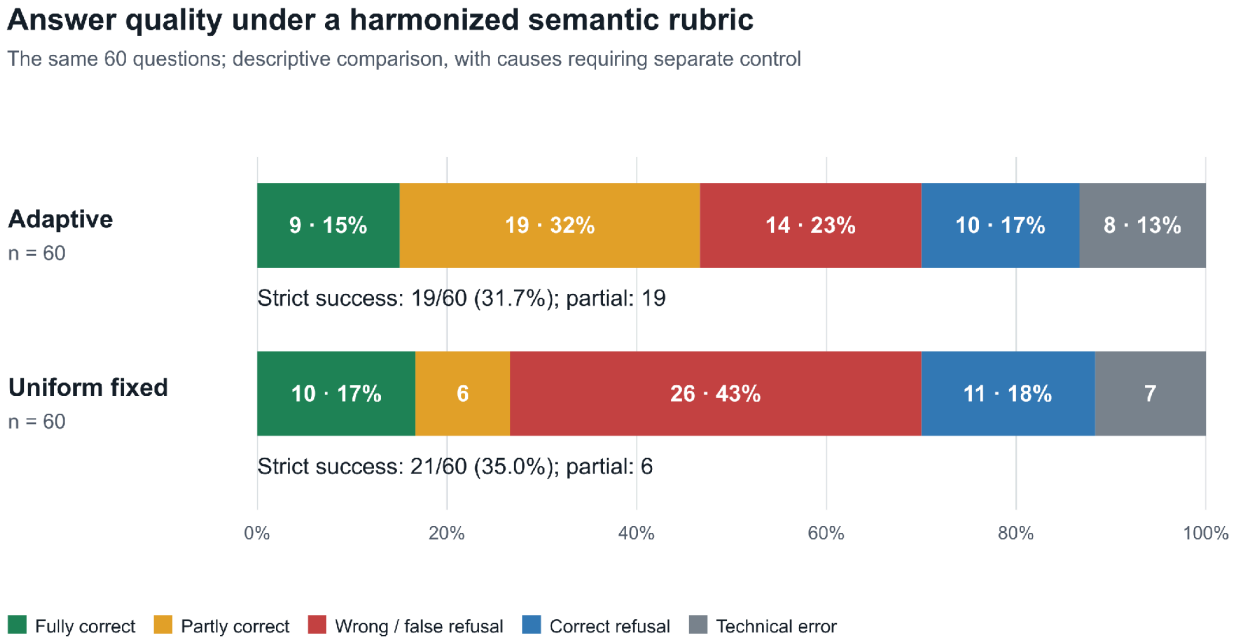


Figure 1 Semantic quality on the same 60 questions in each mode

Note: Strict success and a correct core represent different requirements: a mixed answer with a correct central claim remains partially correct. Similar strict totals do not imply identical error profiles or establish that one mode is superior. All questions and experiments in this study are based on the YouTube video “Subagents vs Agent Teams? Hermes Bots, Goal Loops & Kanban Graphs” (length 42:55; URL: https://www.youtube.com/watch?v=GIW4N7p5KS8&t=6s)[cite: 2, 8, 16].

1 Research problem

Searching a long video combines evidence retrieval, understanding individual frames, interpreting relationships, and temporal localization. An overall success rate mixes these operations with abstentions and execution failures. For an agent with a limited frame budget, this makes diagnosis and the selection of improvements difficult.

Aim and scope

The aim is to determine which video-analysis tasks the VideoAgent v6 configuration with Qwen3-VL-2B-Instruct can perform and to identify observed reasons for failure. The study examines saved results under three conditions on the same set of 60 questions: adaptive, uniform before the fixes, and uniform after the fixes. The analysis covers semantic correctness, temporal localization, citation support, abstention when information is absent, and computational cost.

Research question	Assessment approach
RQ1 How accurate are answers at a known time?	Full and partial correctness; correct core; whether the selected frame falls within the requested second or interval.
RQ2 How correct is reasoning about the video?	Causal relationships and diagram branches; availability of the relevant clip among observations; consistency between conclusions and available evidence.
RQ3 Can the agent identify event onset?	First spoken and visual occurrences; coverage of reference intervals; limitations of the automatic temporal metric.
RQ4 How reliable is answer evaluation?	Agreement between the automatic scorer and semantic review; TP, FP, FN, TN; abstentions, incorrect statuses, and technical failures.
RQ5 How do the modes and fixes differ?	Paired comparison of 60 questions for adaptive and fixed uniform; separate analysis of restored execution in the original uniform implementation.

Analytical approach

The analysis examines whether questions with a known time retain their advantage under strict semantic review, how reasoning depends on access to the relevant clip, and whether a low automatic temporal error indicates correct onset detection. These questions were formulated while analyzing completed runs and are not preregistered hypotheses.

The performance of a vision-language model (VLM) within an agent is distinguished from its ability to understand a clip known to be relevant. Retrieval, formatting, and memory failures are not attributed entirely to the model's reasoning ability. A separate controlled experiment is required to disentangle these factors causally.

2 Materials and methods

The study uses secondary analysis of two audit archives: audit_simple60_v6_run001_2026-10-23 and audit_simple60_uniform_fixed_run001_2026-10-23. The first contains adaptive 60 and original uniform 60; the second contains corrected uniform 60. The complete objects for all 60 questions were verified as identical across the three conditions [1–4]. The simple24_ prefix in some IDs denotes inherited questions within the simple60 set.

Configuration	Adaptive 60	Uniform fixed 60
Model	Qwen3-VL-2B-Instruct	Same model and revision
Model revision	89644892e4d85e24eaac8bacfd4f463576704203	Same
Questions in each of the four task types	15	15
Tool and frame limits	4 calls; 16 frames	4 calls; 16 frames
Image size	560 px	560 px
Maximum new tokens	768 in the configuration	256 observation; 384 final
Fixed uniform context limit	Not compared	6144 tokens
Cache	Disabled	Disabled

Analysis procedure

The unit of analysis is a question’s final result, its citations, inspected frames, and action trace. The review_findings files, specialized temporal and reasoning reviews, predictions, and metrics were compared. Dataset composition, question identity, and metric consistency were checked first. Verdicts were then harmonized under one rubric, followed by calculation of outcome distributions, confusion matrices, time, and resource use. No new answers were generated.

All 60 questions were compared as pairs for the two main modes. Original uniform is analyzed separately because 55/60 results fail technically, preventing a semantic comparison under normal model operation. Differences in code and generation limits remain a limitation of the comparison.

Semantic coding was performed by the assistant using available local evidence; independent inter-rater agreement was not measured. Spoken reference points use automatic speech recognition (ASR) without word-level alignment to the original audio. The checked source-file hashes matched, and the saved automatic metrics equaled those recomputed in the corresponding audits.

3 Results by question type

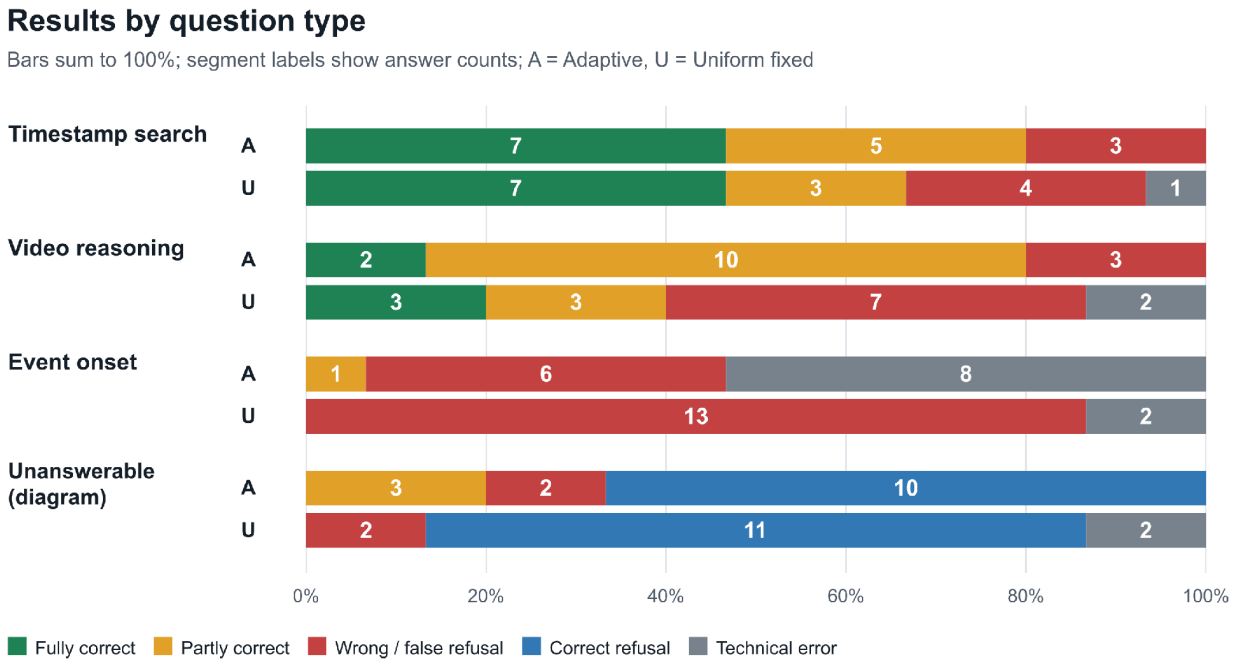


Figure 2 Full quality distribution within each task on identical question sets

Task	Mode	Strict success	Partial	Incorrect	Technical failure
Timestamp search	Adaptive 60	7/15	5	3	0
Timestamp search	Uniform fixed 60	7/15	3	4	1
Video reasoning	Adaptive 60	2/15	10	3	0
Video reasoning	Uniform fixed 60	3/15	3	7	2
Event onset	Adaptive 60	0/15	1	6	8
Event onset	Uniform fixed 60	0/15	0	13	2
Unanswerable	Adaptive 60	10/15	3	2	0
Unanswerable	Uniform fixed 60	11/15	0	2	2

Adaptive provides a correct core in 12/15 timestamp questions, 12/15 reasoning questions, and one onset question. For fixed uniform, the corresponding counts are 10/15, 6/15, and 0/15. Fully correct answers are substantially less common: the central claim is often accompanied by false details.

Fixed uniform has two more strict successes but also more incorrect results: 26 versus 14. Adaptive more often retains a partially useful answer: 19 versus 6. The findings therefore cannot be reduced to a single ranking of modes without specifying requirements for completeness and evidential support.

4 Finding and reading a specified moment

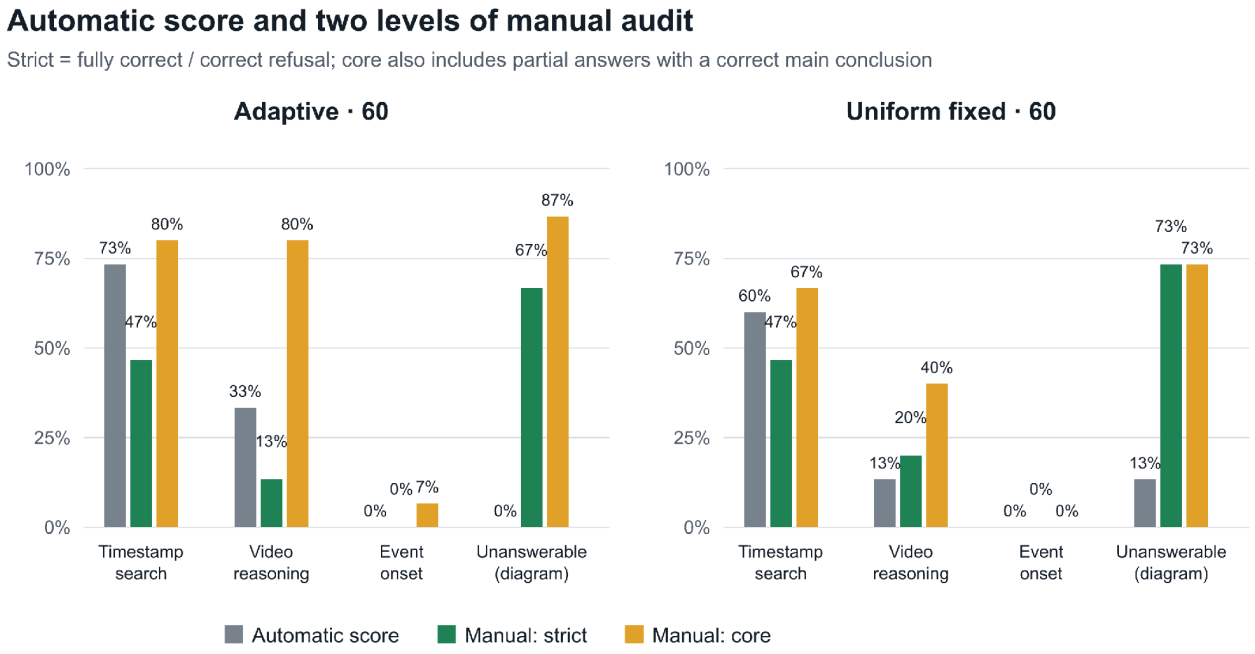


Figure 3 Automatic success strict semantic success and correct core by task

On timestamp questions, the core answer is correct in 12/15 adaptive cases (80.0%) and 10/15 fixed uniform cases (66.7%). Under the adopted rubric, each mode has 7/15 fully correct results. The gap between core correctness and strict success involves unsupported quotes, timing details, interface-element names, and invented visual details.

Example	Model answer and factual check
simple24_time_04	Both modes replace librarian with Gema4, although the correct roles are present in the available ASR. This is an interpretation error despite access to the text.
simple24_time_05	Test is correct, but the claim that NAME is filled at 1736.902 s is contradicted by the frame. A correct name does not support the additional timing claim.
simple24_time_06	The 500-word limit is correct. An additional direct quote incorrectly conflates the five search sources with output-document length.
simple60_time_06	Adaptive confuses 102k and 37k despite its own ASR; fixed uniform abstains although the value is available.
simple60_time_08 and 09	In adaptive, the correct core is accompanied by unsupported interface details; these answers remain partially correct.

In fixed uniform, seven questions with an explicit second received a frame with an error below 0.016 s and a correct core answer. Among the other eight questions about intervals, the core content is correct in only 3/8. This distinguishes the accuracy of frame selection from the accuracy of understanding the retrieved material [2].

5 Video reasoning and evidence availability

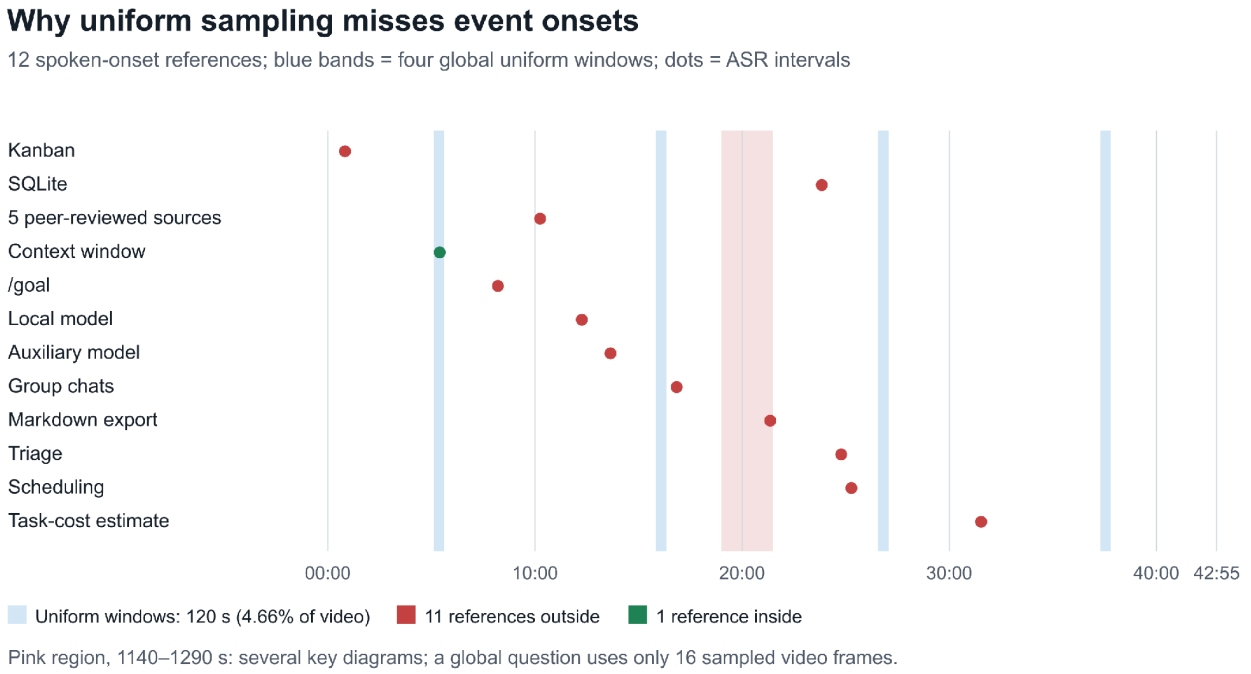


Figure 4 Global fixed uniform windows and events on the video timeline

The core reasoning answer is correct in 12/15 adaptive cases and 6/15 fixed uniform cases. Only 2/15 and 3/15, respectively, are fully correct and clearly grounded. Adaptive thus more often retains a useful core, while both modes show a large gap between the central claim and a fully correct explanation.

Ten adaptive reasoning answers are partially correct: eight mix correct and incorrect details, one has correct content but an incorrect insufficient_evidence status and no citations, and one contains an error in its rationale. Three results are incorrect or unnecessary abstentions. In fixed uniform, three answers are partially correct, seven are incorrect or fail to answer the question, and two are lost to formatting failures.

For 13 global reasoning questions in fixed uniform, the required named diagrams are absent from the inspected windows in ten cases. The main diagram block around 1140–1290 s lies between the windows. These errors cannot all be attributed to the VLM's inability to reason: some arise because the relevant material is missing from the observations.

Interpretation errors also remain when the diagram is available. In adaptive, a request for changes in Email to Decision Brief is incorrectly routed to Approved → Send & save instead of Revise → Brief + draft reply. In A Goal Within a Larger Graph, the Fail and Pass branches are confused. An Oracle Clip experiment is needed to assess understanding separately: the same question on a verified relevant clip under identical frame and generation limits.

6 Detecting the first occurrence of an event

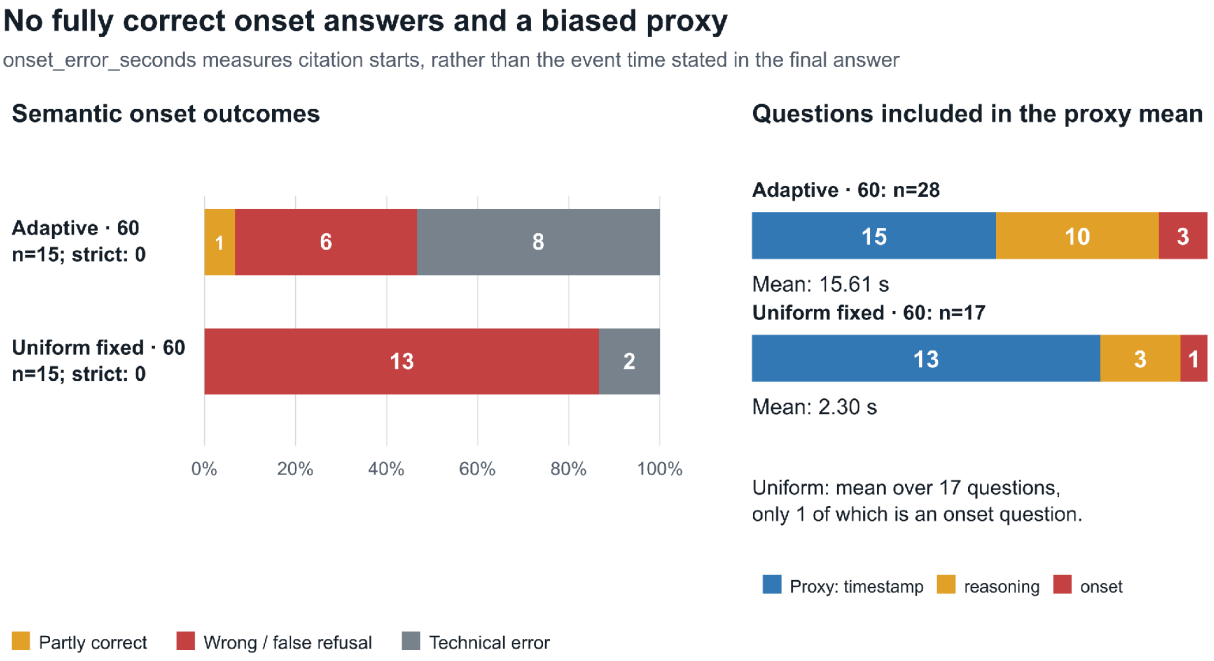


Figure 5 Event-onset answer quality and composition of the automatic temporal proxy

Neither mode produces a fully correct onset answer: 0/15 in each. Adaptive has eight technical failures, six incorrect answers, and one partially correct answer; fixed uniform has two technical failures and 13 incorrect answers. Fewer formatting failures did not produce correct onset results.

Event	Local reference	Observed error
Kanban in speech	47.28–52.48 s	Adaptive uses the start of a much later topical clip around 1441 s.
SQLite in speech	1429.28–1432.24 s	Adaptive gives 1429.695 s but misidentifies the table row; the text appears on screen earlier.
Email to Decision Brief	1150.049–1155.054 s	Fixed uniform sees the slide at 1154.987 s and then denies its appearance.
Two Loops	1235.034–1240.039 s	Fixed uniform sees the slide at 1245.010 s but incorrectly places this time before 20:00.
Task Graphs	1610.042–1615.047 s	Fixed uniform gives 2559.991 s, where the presenter and Subscribe appear.

In fixed uniform, only one of the 12 spoken reference intervals overlaps the four global windows; the other 11 lie outside the observations. This is a coverage measure, not accuracy. Identifying the first visual appearance requires a last-absent frame, a first-present frame, local boundary refinement, and checks for earlier candidates.

The automatic onset_error_seconds value for adaptive is 15.6 s across 28 questions: 15 timestamp, 10 reasoning, and only 3 onset. For fixed uniform it is 2.30 s across 17 questions, only one of which concerns onset. The metric uses the start of an overlapping citation, not the time stated by the model, and excludes non-overlapping errors. A reduction in this conditional mean does not demonstrate improved onset detection.

7 Abstention and the decision to answer

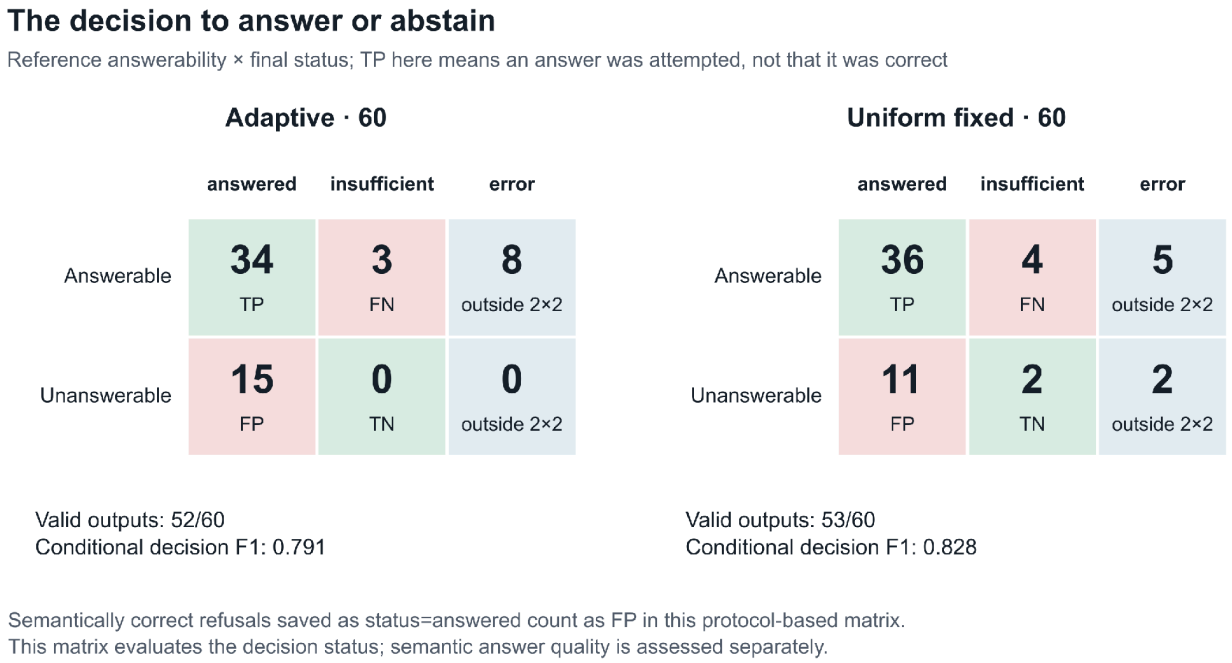


Figure 6 Question answerability and result status with technical failures shown separately

In this matrix, the positive class is an answerable question; a positive decision is status=answered and a negative decision is insufficient_evidence. TP denotes an attempt to answer an answerable question and does not guarantee factual correctness. FP is not equivalent to hallucination: a correct textual abstention can also be stored with answered. Failures are shown in a third column and excluded only from the conditional 2 × 2 matrix.

Measure	Adaptive 60	Uniform fixed 60
TP / FP / FN / TN among valid results	34 / 15 / 3 / 0	36 / 11 / 4 / 2
Precision / Recall	69.4% / 91.9%	76.6% / 90.0%
Specificity / F1	0.0% / 79.1%	15.4% / 82.8%
Balanced accuracy	45.9%	52.7%
Denominator excluding technical failures	52 of 60	53 of 60

There are 10/15 substantively correct, clean abstentions in adaptive and 11/15 in fixed uniform. The correct insufficient_evidence status on unanswerable questions occurs in 0/15 and 2/15. All 15 adaptive unanswerable questions have answered status, including ten clean, correct abstentions; in fixed uniform, nine clean abstentions are also stored with answered.

On unanswerable questions, adaptive additionally gives three partial abstentions with incorrect details or reasoning and two hallucinated answers. Fixed uniform invents an answer twice and loses two final responses to formatting. Substantive abstention, status correctness, and technical validity therefore need separate measurement.

8 Reliability of automatic scoring

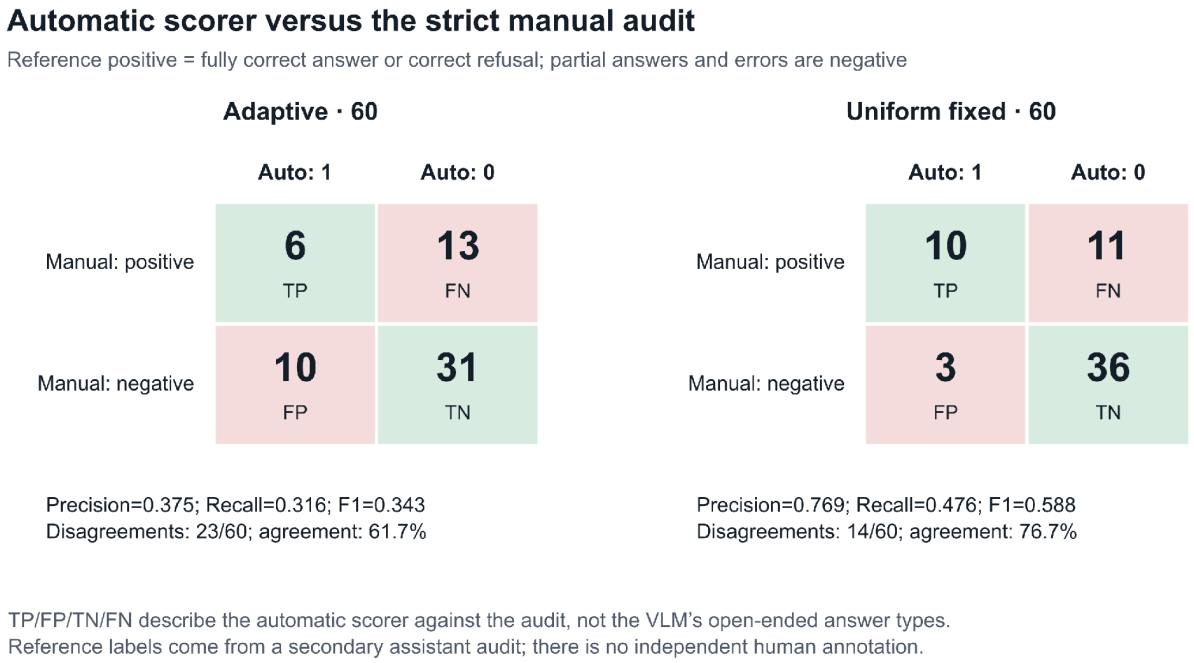


Figure 7 Automatic scorer against the unified strict semantic assessment

The positive reference class is a fully correct semantic outcome, including a correct abstention. A positive prediction is auto_success=1. TP means a correct result is accepted, FP a partially correct or incorrect result is accepted, FN a correct result is rejected, and TN an unsuccessful result is rejected. Partially correct answers are negative under the strict criterion; technical failures remain unsuccessful outcomes.

Measure	Adaptive 60	Uniform fixed 60
TP / FP / FN / TN	6 / 10 / 13 / 31	10 / 3 / 11 / 36
Precision	37.5%	76.9%
Recall	31.6%	47.6%
F1	34.3%	58.8%
Specificity	75.6%	92.3%
Balanced accuracy	53.6%	70.0%
Agreement with semantic review	61.7%	76.7%

The overall automatic score is 16/60 (26.7%) for adaptive and 13/60 (21.7%) for fixed uniform, whereas strict semantic success is 19/60 and 21/60. Lexical scoring accepts some answers with incorrect diagram branches or false additions. Conversely, an incorrect status causes a correct abstention to be rejected. For onset, exact text matching does not replace comparison of timestamps.

Precision and Recall here characterize the scorer, not the VLM's video recognition. The original failing uniform implementation is excluded from these matrices: auto=0 and strict=0 for every question would yield trivial TN=60 and 100% agreement, providing no evidence of model quality.

9 Paired comparison of the same 60 questions

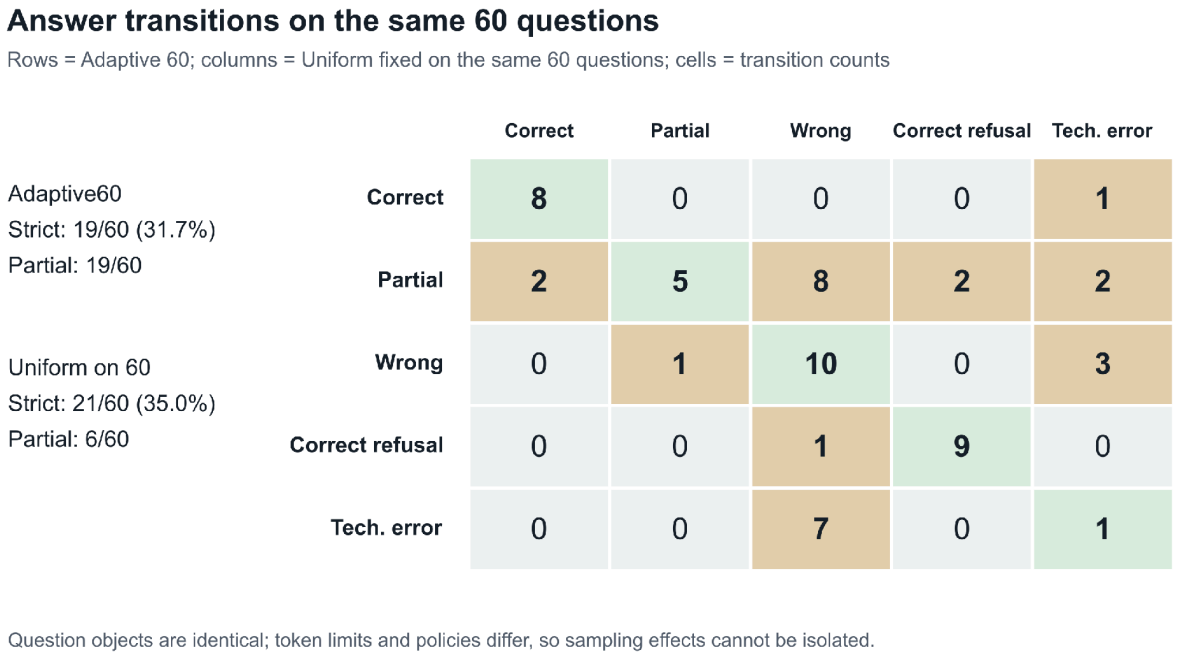


Figure 8 Quality transitions from adaptive to corrected uniform across all 60 questions

Measure	Adaptive 60	Uniform fixed 60
Strict semantic success	19/60 · 31.7%	21/60 · 35.0%
Correct core on answerable questions	25/45 · 55.6%	16/45 · 35.6%
Partially correct answers	19/60	6/60
Incorrect results	14/60	26/60
Technical failures	8/60	7/60
Automatic success	16/60 · 26.7%	13/60 · 21.7%

Seventeen strict successes are retained in both modes; two are lost when moving to fixed uniform and four are gained. The net increase is two results, or 3.3 percentage points. Under the more permissive core criterion across all questions, including partial substantive abstentions, 26 successes are retained, 12 are lost, and one is gained: 38/60 → 27/60. This overall measure must not be confused with 25/45 → 16/45 on answerable questions.

Fixed uniform produces slightly more fully correct outcomes but more often loses partially useful information. A single number cannot establish which overall strategy is better. The comparison is paired by question, but agent code and effective generation limits differ; the causal effect of frame sampling alone is not isolated.

10 Execution time and observation cost

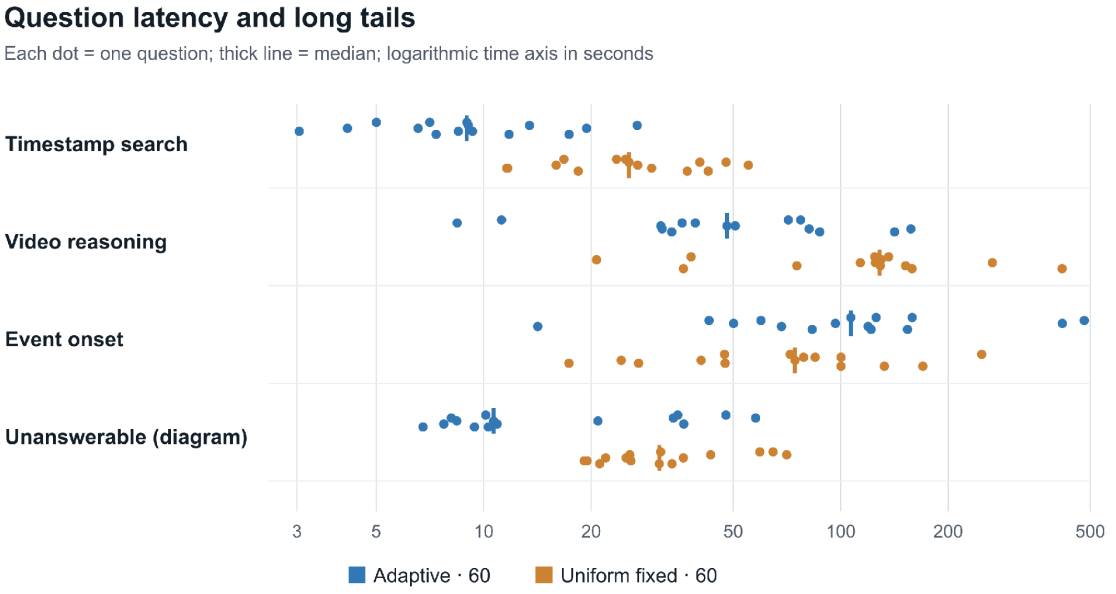


Figure 9 Time per question by task and mode

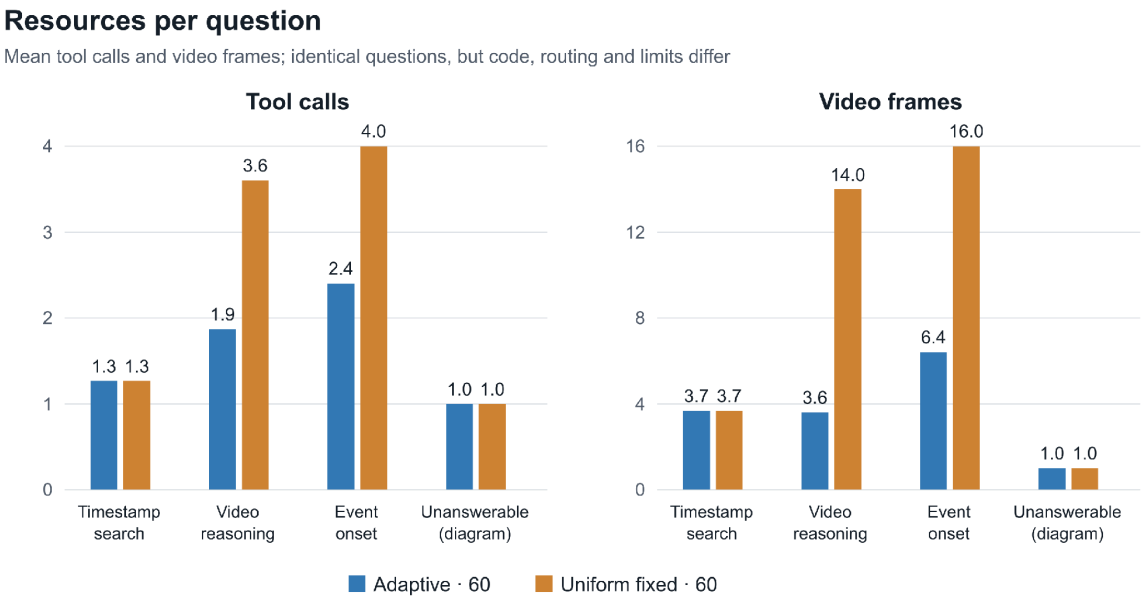


Figure 10 Mean tool calls and frames per question

Measure	Adaptive 60	Uniform fixed 60
Total question time	57 min 57 s	1 h 11 min 11 s
Mean / median question time	57.9 / 32.6 s	71.2 / 41.6 s
90th percentile / maximum	127.2 / 481.0 s	137.8 / 416.7 s
Total tool calls / frames	98 / 220	148 / 520

Question time increased by 22.8%, tool calls by 51.0%, and frames by a factor of 2.36; model calls increased from 208 to 223. Separate wall-clock time for adaptive is unavailable in the original execution containing both modes. Fixed uniform took 72 min 35 s by the clock. Generation accounts for 96.5% of its question time and is partly included in tool time.

11 Final-response failures and invalid retries

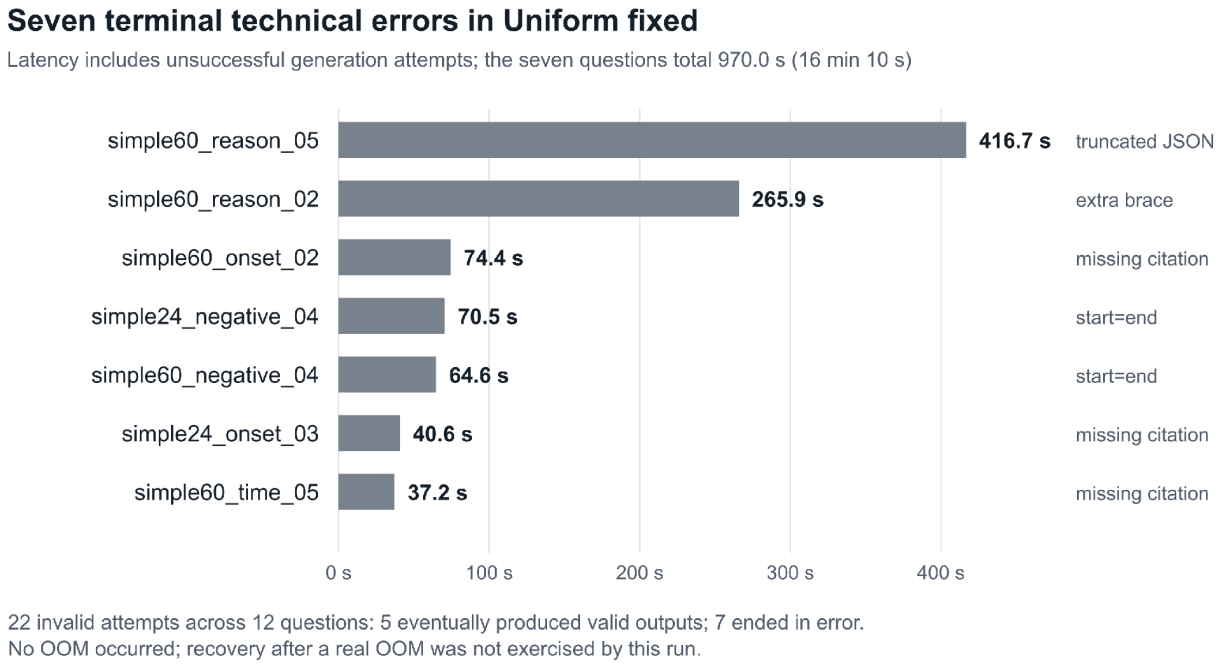


Figure 11 Time spent on the seven fixed uniform questions with a technical final-response failure

Measure	Adaptive 60	Uniform fixed 60
Final-response formatting failures	8/60	7/60
Time spent on questions ending in technical failure	25 min 51 s	16 min 10 s
Share of time spent on failed results	44.6%	22.7%

All eight adaptive failures concern onset. Seven end after three attempts without the required citations to inspected evidence; the remaining one fails the requirement to cite all four windows. Most discarded drafts also contain incorrect times. A draft with the correct numeric timestamp for the introduction of task-cost estimation is likewise lost to formatting and is not counted as a successful result.

Fixed uniform has seven terminal failures: three without citations, two with start=end, one with an extra closing brace following an initial attempt without citations, and one with truncated, repetitive JSON at the 384-token limit. These questions consume 970.0 s, or 22.7% of question time. The most expensive are simple60_reason_05 (416.7 s) and simple60_reason_02 (265.9 s).

Fixed uniform contains 22 invalid attempts across 12 questions: five questions recover and seven are stopped by the repetition guard. Five stops occur after two identical errors, and two after three attempts with a different first error. Successful text generation by the backend does not guarantee a valid final response.

Normalizing citations and statuses must preserve evidential support: an unsupported time interval must not be inserted merely to produce valid JSON. Formatting and content failures are distinct dimensions of quality.

12 Restoring uniform execution

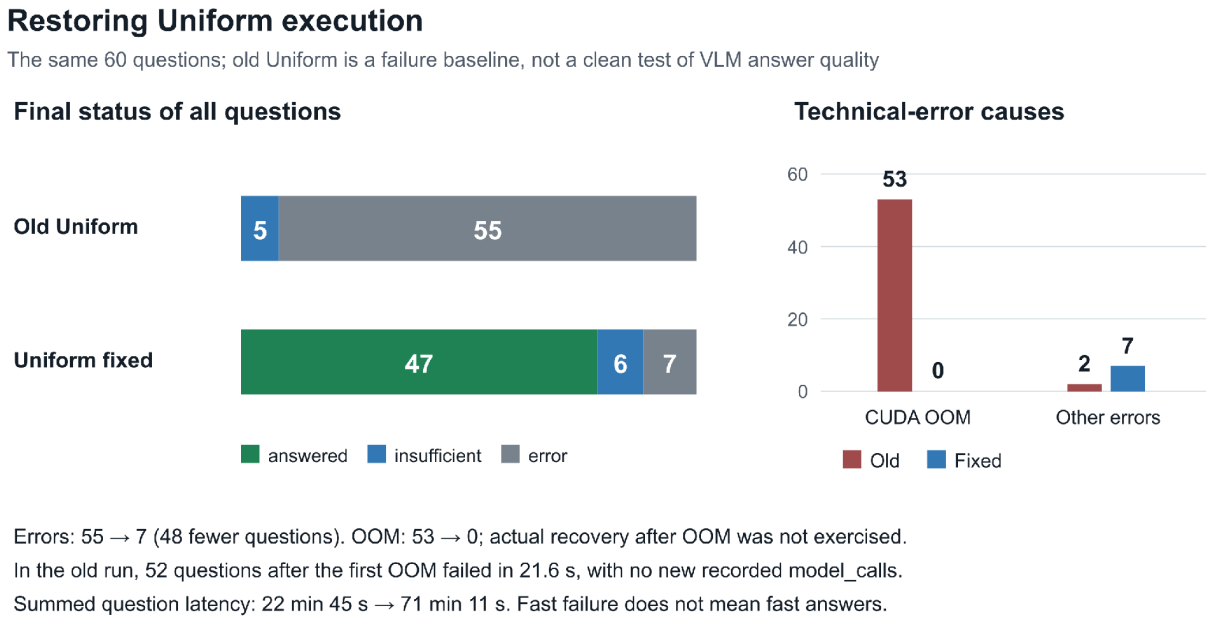


Figure 12 Statuses and technical failures in original and corrected uniform

Measure	Before fixes	After fixes
Valid result by status	5/60	53/60
Answered / insufficient_evidence / error	0 / 5 / 55	47 / 6 / 7
CUDA OOM failures	53	0
Frame-contract failures	2	0 of this kind
Final-response formatting failures	0 as a terminal category	7
Strict semantic success	0/60	21/60

Original uniform is not a valid semantic baseline: 53 results are lost to CUDA OOM, two to frame-contract violations, and the five surviving responses are false abstentions. The first OOM occurs on the eighth question; the following 52 results terminate without an increase in the model_calls counter. The cascade of immediate failures distorts mean and median time.

After the fixes, technical failures decrease from 55/60 to 7/60, a reduction of 48 results. This demonstrates restored execution. It cannot be interpreted as a pure improvement in the VLM's ability to understand video: the original mode was mostly unable to produce an answer. Failure handling, uniform policy, and inference limits all changed.

No OOM occurs in the corrected run, so the actual recovery branch after an OOM is not exercised by this run. The data establish the absence of a new OOM in this run and a substantial reduction in technical failures, but provide no separate estimate of recovery success after an OOM actually occurs.

13 Discussion and conclusions

Within VideoAgent v6, the model is most reliable when reading a specified moment. Adaptive more often provides a correct core on answerable questions, while fixed uniform has two more fully correct outcomes because of a different response profile. Neither mode demonstrates reliable detection of the first occurrence of an event.

The study identifies three distinct limitations. First, sparse global sampling can omit the relevant diagram or early event. Second, even a retrieved diagram may be interpreted with confused branches and unsupported details. Third, a substantively useful answer can be lost because of formatting or an incorrect status. Improving one layer alone does not guarantee improvement across all measures.

Next experiments

Separating retrieval from understanding

Compare Oracle Clip with the regular agent on the same questions under identical frame, context, and generation limits.

Onset search as boundary refinement

Distinguish the first spoken mention from the first visual appearance; retain the last-absent and first-present frames and check earlier candidates.

Evaluating time stated in the final answer

Extract the stated timestamp and measure its distance to the reference interval, hits at tolerances of 1, 5, and 10 s, missed events, and false events with a complete denominator.

Aligning content and output contracts

Measure factual correctness, citation support, status correctness, and final-response validity separately. Retain technical failures in overall success.

Controlled comparison of modes

Fix all limits and code, add multiple videos, and obtain independent coding of disputed cases. Differences observed on one video do not establish general superiority of a mode.

Sources and reproducibility

[1] audit_simple60_v6_run001_2026-10-23: review_findings.json, temporal_review.json, reasoning_review.json, scoring_audit.json, and the HTML review. [2] audit_simple60_uniform_fixed_run001_2026-10-23: corresponding fixed uniform audit files.

[3] VideoAgent/results/simple60_v6/run_001: adaptive and uniform predictions/metrics, manifest, and questions. [4] VideoAgent/results/simple60_uniform_fixed/run_001: uniform predictions/metrics, manifest, and questions. All calculations in this report concern only these two sources; recommendations are not measured improvements.

[5] Subagents vs Agent Teams? 🧠 Hermes Bots, Goal Loops & Kanban Graphs: https://www.youtube.com/watch?v=GlW4N7p5KS8&t=6s

Appendix Adaptive 60 questions 1 to 6

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
24 time_01 Timestamp search	Question: What is the slide title at 19:30? Answer: Email to Decision Brief; lists the diagram stages.	Fully correct Auto: 1; time: 7.4 s Frame 1170.002: Email to Decision Brief.
24 time_02 Timestamp search	Question: What is the final workflow stage shown at 20:20? Answer: Connected library, after Human review and approval.	Fully correct Auto: 1; time: 8.5 s Frame 1219.985: Human review → Approved → Connected library.
24 time_03 Timestamp search	Question: At 26:55, how often does the dispatcher check the queue? Answer: Every 60 seconds.	Fully correct Auto: 1; time: 6.6 s Frame 1615.013: The Dispatcher (60s loop); checks every 60 seconds.
24 time_04 Timestamp search	Question: Which three bot roles are named between 17:15 and 17:25? Answer: Orchestrator, Researcher, Gema4; replaces Librarian.	Incorrect Auto: 0; time: 13.4 s ASR 1035.65–1042.81: librarian, orchestrator, researcher.
24 time_05 Timestamp search	Question: What name is given to the new board between 28:55 and 29:10? Answer: Test; incorrectly claims the NAME field is already filled at 1736.902 s.	Partially correct Auto: 1; time: 7.1 s ASR 1739.38–1744.54 names test; the NAME field is still empty in frame 1736.902.
24 time_06 Timestamp search	Question: What word limit is set between 31:50 and 32:20? Answer: 500 words; invents a quotation limiting the search to 500 words instead of five sources.	Partially correct Auto: 1; time: 11.8 s ASR 1914.17–1918.25 and frames 1923/1927: five sources for the search; output document limited to 500 words.

Fully correct means semantically correct under the adopted codebook. Partially correct means a correct component with errors or incomplete support. Correct abstention is assessed semantically, regardless of status. Technical failure is not counted as success even when the draft is correct.

Appendix Adaptive 60 questions 7 to 12

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
60 time_01 Timestamp search	Question: At 00:20, what black object is on the desk? Answer: A black microphone on the desk.	Fully correct Auto: 1; time: 3.0 s Frame 19.987: a black microphone on the desk.
60 time_02 Timestamp search	Question: At 19:00, what is the middle column called? Answer: The middle column is Working.	Fully correct Auto: 1; time: 4.2 s Frame 1140.006: the middle column is Working.
60 time_03 Timestamp search	Question: At 21:20, which feature is being added? Answer: Add Markdown export; imprecisely describes the diagram title as a Goal box label.	Fully correct Auto: 1; time: 9.1 s Frame 1280.012: Your task → Add Markdown export.
60 time_04 Timestamp search	Question: At 23:00, which two systems does the table compare? Answer: Kanban and delegated_task; a typo in delegate_task.	Fully correct Auto: 0; time: 5.0 s Table at 1380: Kanban and delegate_task.
60 time_05 Timestamp search	Question: Which tool does he enable between 02:50 and 03:10? Answer: Task Delegation; adds an unverified exact activation time of 181.248 s.	Partially correct Auto: 1; time: 17.3 s ASR T00043–T00044 and saved evidence E001/E002: Task Delegation.
60 time_06 Timestamp search	Question: Between 05:45 and 06:00, what token count is shown for the delegating parent? Answer: Claims the count is not visible; says the parent used 102000 tokens instead of 37000.	Incorrect Auto: 0; time: 9.3 s Saved E001 / ASR: 37,000 tokens for the parent agent, excluding subagent usage.

Fully correct means semantically correct under the adopted codebook. Partially correct means a correct component with errors or incomplete support. Correct abstention is assessed semantically, regardless of status. Technical failure is not counted as success even when the draft is correct.

Appendix Adaptive 60 questions 13 to 18

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
60 time_07 Timestamp search	Question: Between 11:40 and 12:00, what can a long-running goal exhaust? Answer: A long-running goal exhausts the user's reasoning/thought resources rather than the model usage quota.	Incorrect Auto: 0; time: 19.4 s ASR 711.31–714.15: your quota is consumed.
60 time_08 Timestamp search	Question: Between 30:00 and 30:20, which profile assigns tasks to agents? Answer: Orchestrator; invents a sequence of selected profiles while the checkmark remains on default.	Partially correct Auto: 1; time: 26.9 s ASR 1802.54–1814.34: orchestrator assigns tasks; the menu checkmark stays on default.
60 time_09 Timestamp search	Question: What benefit of automatic decomposition is named between 39:30 and 39:40? Answer: Starts with Auto-decompose tasks and an invented toggle, but correctly quotes the convenience of entering a task and clicking go.	Partially correct Auto: 1; time: 9.0 s ASR T00832–T00834: convenience—enter a task, click go, and Hermes decomposes it.
24 reason_01 Video reasoning	Question: In "Email to Decision Brief," what happens if changes are requested? Answer: Requested changes are approved and saved; confuses Approved with Revise.	Incorrect Auto: 1; time: 33.6 s Diagram at 1170: Human review → Revise → Brief + draft reply; Approved → Send & save.
24 reason_02 Video reasoning	Question: In "Article to Curated Knowledge," can an entry reach the library without approval? Answer: Abstains: insufficient evidence to establish whether approval is required.	Incorrect Auto: 0; time: 157.1 s Diagram at 1220 and ASR 1230.73–1235.13: an approved entry is added to the library.
24 reason_03 Video reasoning	Question: In "Two Loops," what happens when evidence is missing? Answer: Return to research / bounded repair, but incorrectly assigns human note-taking to the researcher.	Partially correct Auto: 1; time: 48.0 s Diagram at 1250: Source checks → Missing evidence / bounded repair → Researcher.

Fully correct means semantically correct under the adopted codebook. Partially correct means a correct component with errors or incomplete support. Correct abstention is assessed semantically, regardless of status. Technical failure is not counted as success even when the draft is correct.

Appendix Adaptive 60 questions 19 to 24

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
24 reason_04 Video reasoning	Question: In "Two Loops," where does a draft go for revision? Answer: Includes the correct back to Librarian, but invents Draft an entry → Curation Loop.	Partially correct Auto: 1; time: 39.2 s Diagram at 1250: You / Review entry → Revise → Librarian.
24 reason_05 Video reasoning	Question: In "A Goal Within a Larger Graph," what happens when checks fail? Answer: Repair/recheck until completion; incorrectly proposes Reviewer rather than Implementer after continued failure.	Partially correct Auto: 0; time: 77.2 s Diagram at 1280, also visible around 1297–1313: Fail → Implementer; Pass → Reviewer; budget reached → Orchestrator.
24 reason_06 Video reasoning	Question: In "How Task Graphs Actually Run," why does a blocked task wait? Answer: Waiting for preceding research work; incorrectly treats SCHEDULED/RUNNING as blocked.	Partially correct Auto: 0; time: 87.4 s Diagram at 1615: dispatcher verifies dependencies; ASR 2003.33–2007.41 explains waiting for preceding research work.
60 reason_01 Video reasoning	Question: In "Email to Decision Brief," what must be prepared between research and human review? Answer: Brief + draft reply / Cited recommendation; correct content saved as insufficient_evidence without citations.	Partially correct Auto: 0; time: 81.6 s Diagram at 1170: Brief + draft reply / cited recommendation between research and Human review.
60 reason_02 Video reasoning	Question: In "Article to Curated Knowledge," why check context and connections before drafting? Answer: Checking context and connections establishes relevance and links to the existing knowledge base.	Fully correct Auto: 0; time: 31.3 s Diagram at 1220: Find the fit / Check context + connections before Draft an entry.
60 reason_03 Video reasoning	Question: In "Two Loops," should blocked work continue automatically? Answer: Correct No, but replaces Pause and ask you with the researcher adding notes.	Partially correct Auto: 0; time: 35.9 s Diagram at 1250: Blocked? Pause and ask you; the dispatcher runs automatically every 60 seconds.

Fully correct means semantically correct under the adopted codebook. Partially correct means a correct component with errors or incomplete support. Correct abstention is assessed semantically, regardless of status. Technical failure is not counted as success even when the draft is correct.

Appendix Adaptive 60 questions 25 to 30

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
60 reason_04 Video reasoning	Question: Why can a well-scoped worker task use a cheaper model? Answer: A clear scope enables a cheaper model; the explanation is tautological and cites irrelevant Vision settings.	Partially correct Auto: 0; time: 71.3 s ASR 307.44–331.60 supports lower intelligence / saving parent context; the full worker-versus-decomposition argument is in ASR 1663.34–1691.66.
60 reason_05 Video reasoning	Question: In "One Workflow, Two Views," is checking separate from the build task? Answer: Correct No: Build & check are combined; adds an irrelevant account of researcher and LLM wiki.	Partially correct Auto: 1; time: 141.5 s Diagram at 1140: Checks is internal to task #2, Build & check export.
60 reason_06 Video reasoning	Question: In "One Workflow, Two Views," which comes first: independent review or whole-outcome review? Answer: Whole-outcome review comes first; reverses the order.	Incorrect Auto: 0; time: 31.6 s Diagram at 1140: Independent review #3, then Whole outcome review #4.
60 reason_07 Video reasoning	Question: In "How Task Graphs Actually Run," are raw ideas sent directly to workers? Answer: Correct No, but substitutes triage → dispatcher → auto-decomposer; incorrectly denies that the dispatcher assigns work to workers.	Partially correct Auto: 1; time: 50.6 s Reference frame 1620.619: Raw Idea → Decomposer → Task Cards → Dispatcher → Worker Nodes.
60 reason_08 Video reasoning	Question: At 36:00, is "Multi-agent research" already finished? Answer: Multi-agent research is unfinished; incorrectly uses badge 1 as evidence of non-completion.	Partially correct Auto: 0; time: 8.4 s Frame 2159.991: Multi-agent research is in READY, not DONE.
60 reason_09 Video reasoning	Question: At 36:00, did the final recommendation stay within 500 words? Answer: The recommendation in DONE stayed within 500 words; does not repeat 485, but correctly affirms compliance.	Fully correct Auto: 0; time: 11.2 s Card in DONE at 2160: standalone 485-word recommendation; 485 ≤ 500.

Fully correct means semantically correct under the adopted codebook. Partially correct means a correct component with errors or incomplete support. Correct abstention is assessed semantically, regardless of status. Technical failure is not counted as success even when the draft is correct.

Appendix Adaptive 60 questions 31 to 36

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
24 onset_01 Event onset	Question: When is Kanban first mentioned? Answer: No valid final response. No valid final response; three drafts giving 1441.2 s without a citation.	Technical failure Auto: 0; time: 121.6 s First matching Kanban ASR cue: 47.28–52.48 s; exact word onset was not aligned to the audio.
24 onset_02 Event onset	Question: When is SQLite first mentioned? Answer: 1429.695 s in the transcript; confuses the Audit trail row with Agents per task.	Partially correct Auto: 0; time: 14.2 s First SQLite ASR cue: 1429.28–1432.24 s; SQLite is already visible around 1354.987 in the Audit trail row. The modality is ambiguous.
24 onset_03 Event onset	Question: When does the presenter first require five peer-reviewed sources? Answer: 602.73 s, the start of the retrieved clip, about 10 s before the event.	Incorrect Auto: 0; time: 50.1 s First ASR cue requiring five peer-reviewed sources: 612.71–617.19 s.
24 onset_04 Event onset	Question: After 19:00, when does the “Email to Decision Brief” slide appear? Answer: No valid final response. No valid final response; draft gives 1140 s instead of the transition around 1150–1155 s.	Technical failure Auto: 0; time: 481.0 s Frames 1150.049 and 1155.054 bracket the transition; the slide is already present in checked frame 1154.987.
24 onset_05 Event onset	Question: After 20:00, when does the “Two Loops” slide appear? Answer: Incorrectly denies that Two Loops appears in any evidence.	Incorrect Auto: 0; time: 158.5 s Frames 1235.034 and 1240.039 bracket the transition; Two Loops is already present in checked frame 1245.010.
24 onset_06 Event onset	Question: After 25:00, when does “How Task Graphs Actually Run” appear? Answer: 25:59.991; invents visual support and incorrectly converts 2559.991 seconds.	Incorrect Auto: 0; time: 153.9 s Reference interval 1610.042–1615.047; local inspection narrows the transition to approximately 1613.5–1613.75.

Fully correct means semantically correct under the adopted codebook. Partially correct means a correct component with errors or incomplete support. Correct abstention is assessed semantically, regardless of status. Technical failure is not counted as success even when the draft is correct.

Appendix Adaptive 60 questions 37 to 42

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
60 onset_01 Event onset	Question: When does he first mention the context window? Answer: No valid final response. No valid final response; draft gives 1410.56 s rather than the first context window mention around 322 s.	Technical failure Auto: 0; time: 119.3 s First spoken context window ASR cue: 321.96–326.92; Context Usage at 310.644 is not a spoken mention.
60 onset_02 Event onset	Question: At what point does he introduce the /goal command? Answer: No valid final response. No valid final response; draft gives 499.566 s, a relevant later /goal frame after the spoken introduction.	Technical failure Auto: 0; time: 96.6 s The /goal command is introduced in ASR 490.49–495.37.
60 onset_03 Event onset	Question: When does he first suggest using a local model? Answer: 815.281 s, a later auxiliary models scene, instead of the first local model suggestion.	Incorrect Auto: 0; time: 42.8 s First suggestion to use a local model: ASR 734.75–737.03.
60 onset_04 Event onset	Question: When does he first mention an auxiliary model? Answer: No valid final response. No valid final response; draft gives 811.91 s, the clip start, not a verified first mention.	Technical failure Auto: 0; time: 106.7 s First auxiliary model ASR cue: 817.85–819.95; the heading is visible earlier, around 815.281. The modality is ambiguous.
60 onset_05 Event onset	Question: When does he start talking about group chats? Answer: No valid final response. No valid final response; draft gives 1110.17 s rather than the first group chats mention around 1008 s.	Technical failure Auto: 0; time: 83.2 s First group chats ASR cue: 1008.15–1012.87.
60 onset_06 Event onset	Question: When does he first mention adding Markdown export? Answer: 1140.0 s; Markdown export is visible, but its first spoken mention is around 1279 s and first visual onset is not established.	Incorrect Auto: 0; time: 68.2 s First spoken Markdown export ASR cue: 1279.45–1283.85; the feature is already visible at 1140. The modality is ambiguous.

Fully correct means semantically correct under the adopted codebook. Partially correct means a correct component with errors or incomplete support. Correct abstention is assessed semantically, regardless of status. Technical failure is not counted as success even when the draft is correct.

Appendix Adaptive 60 questions 43 to 48

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
60 onset_07 Event onset	Question: When does he first describe Triage? Answer: No valid final response. No valid final response; draft gives 1444.810 s and incorrectly describes Triage as a function call.	Technical failure Auto: 0; time: 417.4 s First matching ASR description of Triage: 1486.28–1488.04.
60 onset_08 Event onset	Question: When does he first explain scheduling a task? Answer: 1440.0 s; misattributes a long Triage quotation to another clip and does not explain scheduling.	Incorrect Auto: 0; time: 59.8 s First explanation of running a task at a specified time: ASR 1514.48–1518.76.
60 onset_09 Event onset	Question: When does he introduce the task-cost estimation feature? Answer: No valid final response. No valid final response; 1890.41 matches the ASR cue start, but is described as approximately 30 seconds into the video.	Technical failure Auto: 0; time: 125.6 s Introduction of task-cost estimation: ASR 1890.41–1894.33.
24 negative_01 Unanswerable	Question: On the diagram at 19:30, what is the recipient's email address? Answer: The recipient address is not visible in the diagram.	Correct abstention Auto: 0; time: 6.8 s The diagram at 1170 has no recipient email address; this conclusion applies only to the specified diagram.
24 negative_02 Unanswerable	Question: On the diagram at 20:20, which journal published the article? Answer: The journal name cannot be determined.	Correct abstention Auto: 0; time: 7.7 s The diagram at 1220 has no journal or publisher name.
24 negative_03 Unanswerable	Question: On the diagram at 20:20, how many minutes does human review take? Answer: Human review duration is not specified.	Correct abstention Auto: 0; time: 8.4 s The diagram at 1220 includes Human review without a duration in minutes.

Fully correct means semantically correct under the adopted codebook. Partially correct means a correct component with errors or incomplete support. Correct abstention is assessed semantically, regardless of status. Technical failure is not counted as success even when the draft is correct.

Appendix Adaptive 60 questions 49 to 54

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
24 negative_04 Unanswerable	Question: On the diagram at 20:50, what is the library's total document count? Answer: 1000 documents—an invented count.	Incorrect Auto: 0; time: 34.0 s The diagram at 1250 includes Library without a document count.
24 negative_05 Unanswerable	Question: On the diagram at 21:20, what exact token budget is set for repairs? Answer: The exact token budget is not specified; the introductory budget statement gives no numerical value.	Correct abstention Auto: 0; time: 34.9 s The diagram at 1280 shows repair within budget / budget reached, without a number or token units.
24 negative_06 Unanswerable	Question: On the diagram at 26:55, which exact LLM model powers the decomposer? Answer: Concludes that the exact model is unknown, but first presents the frontier-level description as an LLM name.	Partially correct Auto: 0; time: 9.4 s In the diagram at 1615, frontier-level judgment describes the decomposer; no exact model name is given.
60 negative_01 Unanswerable	Question: On the diagram at 19:30, what is the email's subject line? Answer: The email subject is not shown.	Correct abstention Auto: 0; time: 10.9 s The diagram at 1170 has no email subject line.
60 negative_02 Unanswerable	Question: On the diagram at 20:20, what is the article's DOI? Answer: The article DOI is not specified.	Correct abstention Auto: 0; time: 20.9 s The diagram at 1220 has no article DOI.
60 negative_03 Unanswerable	Question: On the diagram at 20:50, how many repair attempts are allowed? Answer: The repair attempt count is unspecified; adds the false claim that there is a single loop.	Partially correct Auto: 0; time: 57.8 s Bounded repair at 1250 does not specify a numerical attempt limit.

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Appendix Adaptive 60 questions 55 to 60

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
60 negative_04 Unanswerable	Question: On the diagram at 20:50, which software stores the library? Answer: Calls Two Loops, One Knowledge Workflow the library storage software; it is a diagram title.	Incorrect Auto: 0; time: 47.7 s The diagram at 1250 does not name storage software; Two Loops, One Knowledge Workflow is a diagram title, not a software name.
60 negative_05 Unanswerable	Question: On the diagram at 21:20, which programming language implements Markdown export? Answer: The programming language for Markdown export is not specified.	Correct abstention Auto: 0; time: 10.3 s In the diagram at 1280, Markdown is an export format; the implementation language is not specified.
60 negative_06 Unanswerable	Question: On the diagram at 21:20, what is the planned release date? Answer: The date is unknown because 1280.012 s supposedly does not match 21:20; the rationale is incorrect.	Partially correct Auto: 0; time: 36.3 s The diagram at 1280 has no release date.
60 negative_07 Unanswerable	Question: On the diagram at 19:00, how many minutes will task two take? Answer: Task #2 is Waiting; its duration is not specified.	Correct abstention Auto: 0; time: 10.7 s In the diagram at 1140, task #2 is Waiting without a duration.
60 negative_08 Unanswerable	Question: On the diagram at 26:55, what is the queue database's file path? Answer: The database file path is not specified in the diagram.	Correct abstention Auto: 0; time: 8.1 s The diagram at 1615 has no database file path.
60 negative_09 Unanswerable	Question: On the diagram at 26:55, how much RAM does each worker need? Answer: Worker RAM requirements are not specified in the diagram.	Correct abstention Auto: 0; time: 10.1 s The diagram at 1615 shows Worker Nodes without RAM requirements.

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Appendix Uniform fixed 60 questions 1 to 6

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
24 time_01 Timestamp search	Question: What is the slide title at 19:30? Answer: Email to Decision Brief.	Fully correct Auto: 1; time: 18.4 s Frame 1170.002: Email to Decision Brief.
24 time_02 Timestamp search	Question: What is the final workflow stage shown at 20:20? Answer: Connected library after approval.	Fully correct Auto: 1; time: 29.5 s Frame 1219.985: Human review → Approved → Connected library.
24 time_03 Timestamp search	Question: At 26:55, how often does the dispatcher check the queue? Answer: Every 60 seconds.	Fully correct Auto: 1; time: 27.0 s Frame 1615.013: The Dispatcher (60s loop); checks every 60 seconds.
24 time_04 Timestamp search	Question: Which three bot roles are named between 17:15 and 17:25? Answer: Orchestrator, Researcher, Gema4; replaces Librarian.	Incorrect Auto: 0; time: 40.3 s ASR 1035.65–1042.81: librarian, orchestrator, researcher.
24 time_05 Timestamp search	Question: What name is given to the new board between 28:55 and 29:10? Answer: Test; incorrectly claims NAME is already filled at 1736.902.	Partially correct Auto: 1; time: 24.9 s ASR 1739.38–1744.54 names test; the NAME field is still empty in frame 1736.902.
24 time_06 Timestamp search	Question: What word limit is set between 31:50 and 32:20? Answer: 500 words; invents an exact quotation limiting the search to 500 words.	Partially correct Auto: 1; time: 42.5 s ASR 1914.17–1918.25 and frames 1923/1927: five sources for the search; output document limited to 500 words.

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Appendix Uniform fixed 60 questions 7 to 12

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
60 time_01 Timestamp search	Question: At 00:20, what black object is on the desk? Answer: A black microphone.	Fully correct Auto: 1; time: 11.6 s Frame 19.987: a black microphone on the desk.
60 time_02 Timestamp search	Question: At 19:00, what is the middle column called? Answer: Working.	Fully correct Auto: 1; time: 16.0 s Frame 1140.006: the middle column is Working.
60 time_03 Timestamp search	Question: At 21:20, which feature is being added? Answer: Add Markdown export.	Fully correct Auto: 1; time: 25.5 s Frame 1280.012: Your task → Add Markdown export.
60 time_04 Timestamp search	Question: At 23:00, which two systems does the table compare? Answer: Kanban and delegated_task; a typo in delegate_task.	Fully correct Auto: 0; time: 16.8 s Table at 1380: Kanban and delegate_task.
60 time_05 Timestamp search	Question: Which tool does he enable between 02:50 and 03:10? Answer: No valid final response. No final response; correct Task Delegation draft without citations.	Technical failure Auto: 0; time: 37.2 s ASR T00043–T00044 and saved evidence E001/E002: Task Delegation.
60 time_06 Timestamp search	Question: Between 05:45 and 06:00, what token count is shown for the delegating parent? Answer: Incorrect abstention: claims the token counter is not visible.	Incorrect Auto: 0; time: 11.7 s Saved E001 / ASR: 37,000 tokens for the parent agent, excluding subagent usage.

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Appendix Uniform fixed 60 questions 13 to 18

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
60 time_07 Timestamp search	Question: Between 11:40 and 12:00, what can a long-running goal exhaust? Answer: The user's reasoning/thought resources are exhausted rather than the model quota.	Incorrect Auto: 0; time: 55.1 s ASR 711.31–714.15: your quota is consumed.
60 time_08 Timestamp search	Question: Between 30:00 and 30:20, which profile assigns tasks to agents? Answer: Orchestrator; incorrectly says this profile is consistently selected in the menu.	Partially correct Auto: 1; time: 47.7 s ASR 1802.54–1814.34: orchestrator assigns tasks; the menu checkmark stays on default.
60 time_09 Timestamp search	Question: What benefit of automatic decomposition is named between 39:30 and 39:40? Answer: Auto-decompose tasks repeats the feature name instead of explaining its benefit; invents screen elements.	Incorrect Auto: 0; time: 23.5 s ASR T00832–T00834: convenience—enter a task, click go, and Hermes decomposes it.
24 reason_01 Video reasoning	Question: In "Email to Decision Brief," what happens if changes are requested? Answer: Incorrectly states that the video gives no answer about requested changes.	Incorrect Auto: 0; time: 158.3 s Diagram at 1170: Human review → Revise → Brief + draft reply; Approved → Send & save.
24 reason_02 Video reasoning	Question: In "Article to Curated Knowledge," can an entry reach the library without approval? Answer: Correct No, but invents a Research Team / Approved diagram; the explanation contradicts itself.	Partially correct Auto: 0; time: 152.0 s Diagram at 1220 and ASR 1230.73–1235.13: an approved entry is added to the library.
24 reason_03 Video reasoning	Question: In "Two Loops," what happens when evidence is missing? Answer: Describes its own lack of observations instead of the Missing evidence branch.	Incorrect Auto: 0; time: 113.5 s Diagram at 1250: Source checks → Missing evidence / bounded repair → Researcher.

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Appendix Uniform fixed 60 questions 19 to 24

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
24 reason_04 Video reasoning	Question: In "Two Loops," where does a draft go for revision? Answer: Invents draft as a Context Usage menu item.	Incorrect Auto: 0; time: 129.9 s Diagram at 1250: You / Review entry → Revise → Librarian.
24 reason_05 Video reasoning	Question: In "A Goal Within a Larger Graph," what happens when checks fail? Answer: Unconditionally claims BLOCKED and a new dispatcher cycle after Checks Fail.	Incorrect Auto: 0; time: 124.5 s Diagram at 1280, also visible around 1297–1313: Fail → Implementer; Pass → Reviewer; budget reached → Orchestrator.
24 reason_06 Video reasoning	Question: In "How Task Graphs Actually Run," why does a blocked task wait? Answer: Explains waiting through a missing body and unscheduled status, rather than a dependency on preceding work.	Incorrect Auto: 0; time: 36.3 s Diagram at 1615: dispatcher verifies dependencies; ASR 2003.33–2007.41 explains waiting for preceding research work.
60 reason_01 Video reasoning	Question: In "Email to Decision Brief," what must be prepared between research and human review? Answer: Multi-agent research must be completed; invents a REVIEW → DONE transition.	Incorrect Auto: 0; time: 75.3 s Diagram at 1170: Brief + draft reply / cited recommendation between research and Human review.
60 reason_02 Video reasoning	Question: In "Article to Curated Knowledge," why check context and connections before drafting? Answer: No valid final response. No final response; invalid JSON and an irrelevant draft.	Technical failure Auto: 0; time: 265.9 s Diagram at 1220: Find the fit / Check context + connections before Draft an entry.
60 reason_03 Video reasoning	Question: In "Two Loops," should blocked work continue automatically? Answer: Correct No / human input, but invents manual dispatcher activation.	Partially correct Auto: 0; time: 128.4 s Diagram at 1250: Blocked? Pause and ask you; the dispatcher runs automatically every 60 seconds.

Fully correct means semantically correct under the adopted codebook. Partially correct means a correct component with errors or incomplete support. Correct abstention is assessed semantically, regardless of status. Technical failure is not counted as success even when the draft is correct.

Appendix Uniform fixed 60 questions 25 to 30

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
60 reason_04 Video reasoning	Question: Why can a well-scoped worker task use a cheaper model? Answer: A scoped worker task does not require the expensive model's full capabilities and context; grounding is incomplete.	Partially correct Auto: 0; time: 136.2 s ASR 307.44–331.60 supports lower intelligence / saving parent context; the full worker-versus-decomposition argument is in ASR 1663.34–1691.66.
60 reason_05 Video reasoning	Question: In "One Workflow, Two Views," is checking separate from the build task? Answer: No valid final response. No final response; a repetitive draft is truncated three times by the 384-token limit.	Technical failure Auto: 0; time: 416.7 s Diagram at 1140: Checks is internal to task #2, Build & check export.
60 reason_06 Video reasoning	Question: In "One Workflow, Two Views," which comes first: independent review or whole-outcome review? Answer: Whole-outcome review comes first; reverses the order.	Incorrect Auto: 0; time: 129.1 s Diagram at 1140: Independent review #3, then Whole outcome review #4.
60 reason_07 Video reasoning	Question: In "How Task Graphs Actually Run," are raw ideas sent directly to workers? Answer: No: Raw Idea → Decomposer → Task Cards → Dispatcher → Workers.	Fully correct Auto: 1; time: 125.0 s Sampled frame 1620.619: Raw Idea → Decomposer → Task Cards → Dispatcher → Worker Nodes.
60 reason_08 Video reasoning	Question: At 36:00, is "Multi-agent research" already finished? Answer: No, the task is still READY.	Fully correct Auto: 1; time: 20.7 s Frame 2159.991: Multi-agent research is in READY, not DONE.
60 reason_09 Video reasoning	Question: At 36:00, did the final recommendation stay within 500 words? Answer: Yes, the recommendation stayed within 500 words.	Fully correct Auto: 0; time: 38.1 s Card in DONE at 2160: standalone 485-word recommendation; 485 ≤ 500.

Fully correct means semantically correct under the adopted codebook. Partially correct means a correct component with errors or incomplete support. Correct abstention is assessed semantically, regardless of status. Technical failure is not counted as success even when the draft is correct.

Appendix Uniform fixed 60 questions 31 to 36

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
24 onset_01 Event onset	Question: When is Kanban first mentioned? Answer: 306.88 s, the first uniform window start, instead of the first Kanban mention.	Incorrect Auto: 0; time: 169.8 s First matching Kanban ASR cue: 47.28–52.48 s; exact word onset was not aligned to the audio.
24 onset_02 Event onset	Question: When is SQLite first mentioned? Answer: 333.133 s and several invented SQLite appearances.	Incorrect Auto: 0; time: 27.1 s First SQLite ASR cue: 1429.28–1432.24 s; SQLite is already visible around 1354.987 in the Audit trail row. The modality is ambiguous.
24 onset_03 Event onset	Question: When does the presenter first require five peer-reviewed sources? Answer: No valid final response. No final response; draft gives 333.133 without citations.	Technical failure Auto: 0; time: 40.6 s First ASR cue requiring five peer-reviewed sources: 612.71–617.19 s.
24 onset_04 Event onset	Question: After 19:00, when does the “Email to Decision Brief” slide appear? Answer: Incorrectly denies Email to Decision Brief after 19:00.	Incorrect Auto: 0; time: 100.2 s Frames 1150.049 and 1155.054 bracket the transition; the slide is already present in sampled frame 1154.987.
24 onset_05 Event onset	Question: After 20:00, when does the “Two Loops” slide appear? Answer: Incorrectly denies Two Loops after 20:00, claiming that 1245 s is before 20:00.	Incorrect Auto: 0; time: 248.1 s Frames 1235.034 and 1240.039 bracket the transition; Two Loops is already present in sampled frame 1245.010.
24 onset_06 Event onset	Question: After 25:00, when does “How Task Graphs Actually Run” appear? Answer: 2559.991 s; the source frame shows the presenter and Subscribe.	Incorrect Auto: 0; time: 132.4 s Reference interval 1610.042–1615.047; local inspection narrows the transition to approximately 1613.5–1613.75.

Fully correct means semantically correct under the adopted codebook. Partially correct means a correct component with errors or incomplete support. Correct abstention is assessed semantically, regardless of status. Technical failure is not counted as success even when the draft is correct.

Appendix Uniform fixed 60 questions 37 to 42

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
60 onset_01 Event onset	Question: When does he first mention the context window? Answer: 310.644 s; substitutes the visible Context Usage item for the first spoken mention.	Incorrect Auto: 0; time: 47.4 s First spoken context window ASR cue: 321.96–326.92; Context Usage at 310.644 is not a spoken mention.
60 onset_02 Event onset	Question: At what point does he introduce the /goal command? Answer: No valid final response. No final response; draft gives 333.133 without citations.	Technical failure Auto: 0; time: 74.4 s The /goal command is introduced in ASR 490.49–495.37.
60 onset_03 Event onset	Question: When does he first suggest using a local model? Answer: 954.387 s instead of the first local model suggestion.	Incorrect Auto: 0; time: 78.7 s First suggestion to use a local model: ASR 734.75–737.03.
60 onset_04 Event onset	Question: When does he first mention an auxiliary model? Answer: 333.133 s instead of the first auxiliary model mention.	Incorrect Auto: 0; time: 47.2 s First auxiliary model ASR cue: 817.85–819.95; the heading is visible earlier, around 815.281. The modality is ambiguous.
60 onset_05 Event onset	Question: When does he start talking about group chats? Answer: 333.133, 961.894 and 1620.619 s; none establishes the first group chats mention.	Incorrect Auto: 0; time: 100.1 s First group chats ASR cue: 1008.15–1012.87.
60 onset_06 Event onset	Question: When does he first mention adding Markdown export? Answer: 954.387 s; contradicts itself by citing the absence of Markdown export.	Incorrect Auto: 0; time: 17.3 s First spoken Markdown export ASR cue: 1279.45–1283.85; the feature is already visible at 1140. The modality is ambiguous.

Fully correct means semantically correct under the adopted codebook. Partially correct means a correct component with errors or incomplete support. Correct abstention is assessed semantically, regardless of status. Technical failure is not counted as success even when the draft is correct.

Appendix Uniform fixed 60 questions 43 to 48

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
60 onset_07 Event onset	Question: When does he first describe Triage? Answer: 1598.130 s; a later task card rather than the first Triage description.	Incorrect Auto: 0; time: 24.3 s First matching ASR description of Triage: 1486.28–1488.04.
60 onset_08 Event onset	Question: When does he first explain scheduling a task? Answer: 333.133 s instead of the first scheduling explanation.	Incorrect Auto: 0; time: 84.8 s First explanation of running a task at a specified time: ASR 1514.48–1518.76.
60 onset_09 Event onset	Question: When does he introduce the task-cost estimation feature? Answer: 310.644 and other arbitrary times instead of the first cost estimation introduction.	Incorrect Auto: 0; time: 72.2 s Introduction of task-cost estimation: ASR 1890.41–1894.33.
24 negative_01 Unanswerable	Question: On the diagram at 19:30, what is the recipient's email address? Answer: The recipient address is not visible.	Correct abstention Auto: 0; time: 21.1 s The diagram at 1170 has no recipient email address; this conclusion applies only to the specified diagram.
24 negative_02 Unanswerable	Question: On the diagram at 20:20, which journal published the article? Answer: The journal name is not specified.	Correct abstention Auto: 1; time: 19.1 s The diagram at 1220 has no journal or publisher name.
24 negative_03 Unanswerable	Question: On the diagram at 20:20, how many minutes does human review take? Answer: Human review duration is unknown.	Correct abstention Auto: 1; time: 25.0 s The diagram at 1220 includes Human review without a duration in minutes.

Fully correct means semantically correct under the adopted codebook. Partially correct means a correct component with errors or incomplete support. Correct abstention is assessed semantically, regardless of status. Technical failure is not counted as success even when the draft is correct.

Appendix Uniform fixed 60 questions 49 to 54

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
24 negative_04 Unanswerable	Question: On the diagram at 20:50, what is the library's total document count? Answer: No valid final response. No final response; correct abstention in the draft, but the citation has start=end.	Technical failure Auto: 0; time: 70.5 s The diagram at 1250 includes Library without a document count.
24 negative_05 Unanswerable	Question: On the diagram at 21:20, what exact token budget is set for repairs? Answer: 1000 tokens—an invented number.	Incorrect Auto: 0; time: 59.3 s The diagram at 1280 shows repair within budget / budget reached, without a number or token units.
24 negative_06 Unanswerable	Question: On the diagram at 26:55, which exact LLM model powers the decomposer? Answer: The exact LLM name is not specified; frontier-level is a capability description.	Correct abstention Auto: 0; time: 31.0 s In the diagram at 1615, frontier-level judgment describes the decomposer; no exact model name is given.
60 negative_01 Unanswerable	Question: On the diagram at 19:30, what is the email's subject line? Answer: The email subject is not visible.	Correct abstention Auto: 0; time: 25.8 s The diagram at 1170 has no email subject line.
60 negative_02 Unanswerable	Question: On the diagram at 20:20, what is the article's DOI? Answer: The DOI is not visible.	Correct abstention Auto: 0; time: 36.3 s The diagram at 1220 has no article DOI.
60 negative_03 Unanswerable	Question: On the diagram at 20:50, how many repair attempts are allowed? Answer: 1 attempt—an invented number.	Incorrect Auto: 0; time: 43.2 s Bounded repair at 1250 does not specify a numerical attempt limit.

Fully correct means semantically correct under the adopted codebook. Partially correct means a correct component with errors or incomplete support. Correct abstention is assessed semantically, regardless of status. Technical failure is not counted as success even when the draft is correct.

Appendix Uniform fixed 60 questions 55 to 60

Responses are summarized in shortened substantive form; factual checks are provided separately. The 24 and 60 ID prefixes denote question groups within the same simple60 set. Auto denotes success according to the saved automatic scorer.

ID and task	Question and model response	Outcome and factual check
60 negative_04 Unanswerable	Question: On the diagram at 20:50, which software stores the library? Answer: No valid final response. No final response; the draft mistakes the diagram title for software, and the citation has start=end.	Technical failure Auto: 0; time: 64.6 s The diagram at 1250 does not name storage software; Two Loops, One Knowledge Workflow is a diagram title, not a software name.
60 negative_05 Unanswerable	Question: On the diagram at 21:20, which programming language implements Markdown export? Answer: The implementation language is not specified.	Correct abstention Auto: 0; time: 33.7 s In the diagram at 1280, Markdown is an export format; the implementation language is not specified.
60 negative_06 Unanswerable	Question: On the diagram at 21:20, what is the planned release date? Answer: The release date is not specified.	Correct abstention Auto: 0; time: 19.5 s The diagram at 1280 has no release date.
60 negative_07 Unanswerable	Question: On the diagram at 19:00, how many minutes will task two take? Answer: The second task’s duration is not specified.	Correct abstention Auto: 0; time: 22.0 s In the diagram at 1140, task #2 is Waiting without a duration.
60 negative_08 Unanswerable	Question: On the diagram at 26:55, what is the queue database's file path? Answer: The database path cannot be determined.	Correct abstention Auto: 0; time: 25.7 s The diagram at 1615 has no database file path.
60 negative_09 Unanswerable	Question: On the diagram at 26:55, how much RAM does each worker need? Answer: Worker RAM requirements are not specified.	Correct abstention Auto: 0; time: 31.3 s The diagram at 1615 shows Worker Nodes without RAM requirements.

Fully correct means semantically correct under the adopted codebook. Partially correct means a correct component with errors or incomplete support. Correct abstention is assessed semantically, regardless of status. Technical failure is not counted as success even when the draft is correct.

Appendix Original uniform results 1 to 20

Questions are identical to the two main conditions and can be located by ID in the preceding appendices. This table accounts for all 60 original results without interpreting OOM as a failure to understand the video. No row contains a valid semantic success.

ID	Outcome	Result content	Time
simple24_time_01	False abstention	Incorrect abstention: claims the title at 19:30 is unconfirmed and mentions a different title, Bot Mode & Intro To Graphs.	185.1 s
simple24_time_02	Frame contract	No valid final response was obtained.	377.7 s
simple24_time_03	Frame contract	No valid final response was obtained.	329.6 s
simple24_time_04	False abstention	Abstains: no role information; zero frames were inspected after temporal-scope filtering.	2.2 s
simple24_time_05	False abstention	Abstains: the board name is unknown; zero frames were inspected after scope filtering.	2.0 s
simple24_time_06	False abstention	Abstains: the word limit is unknown; zero frames were inspected after scope filtering.	1.9 s
simple24_reason_01	False abstention	Incorrect abstention: claims the video gives no answer about requested changes.	178.1 s
simple24_reason_02	CUDA OOM	No valid final response was obtained.	266.7 s
simple24_reason_03	CUDA OOM	No valid final response was obtained.	0.6 s
simple24_reason_04	CUDA OOM	No valid final response was obtained.	0.4 s
simple24_reason_05	CUDA OOM	No valid final response was obtained.	0.4 s
simple24_reason_06	CUDA OOM	No valid final response was obtained.	0.4 s
simple24_onset_01	CUDA OOM	No valid final response was obtained.	0.4 s
simple24_onset_02	CUDA OOM	No valid final response was obtained.	0.4 s
simple24_onset_03	CUDA OOM	No valid final response was obtained.	0.5 s
simple24_onset_04	CUDA OOM	No valid final response was obtained.	0.5 s
simple24_onset_05	CUDA OOM	No valid final response was obtained.	0.5 s
simple24_onset_06	CUDA OOM	No valid final response was obtained.	0.5 s
simple24_negative_01	CUDA OOM	No valid final response was obtained.	0.5 s
simple24_negative_02	CUDA OOM	No valid final response was obtained.	0.4 s

Appendix Original uniform results 21 to 40

Questions are identical to the two main conditions and can be located by ID in the preceding appendices. This table accounts for all 60 original results without interpreting OOM as a failure to understand the video. No row contains a valid semantic success.

ID	Outcome	Result content	Time
simple24_negative_03	CUDA OOM	No valid final response was obtained.	0.4 s
simple24_negative_04	CUDA OOM	No valid final response was obtained.	0.4 s
simple24_negative_05	CUDA OOM	No valid final response was obtained.	0.4 s
simple24_negative_06	CUDA OOM	No valid final response was obtained.	0.4 s
simple60_time_01	CUDA OOM	No valid final response was obtained.	0.4 s
simple60_time_02	CUDA OOM	No valid final response was obtained.	0.4 s
simple60_time_03	CUDA OOM	No valid final response was obtained.	0.4 s
simple60_time_04	CUDA OOM	No valid final response was obtained.	0.4 s
simple60_time_05	CUDA OOM	No valid final response was obtained.	0.0 s
simple60_time_06	CUDA OOM	No valid final response was obtained.	0.0 s
simple60_time_07	CUDA OOM	No valid final response was obtained.	0.0 s
simple60_time_08	CUDA OOM	No valid final response was obtained.	0.0 s
simple60_time_09	CUDA OOM	No valid final response was obtained.	0.0 s
simple60_reason_01	CUDA OOM	No valid final response was obtained.	0.4 s
simple60_reason_02	CUDA OOM	No valid final response was obtained.	0.4 s
simple60_reason_03	CUDA OOM	No valid final response was obtained.	0.4 s
simple60_reason_04	CUDA OOM	No valid final response was obtained.	0.4 s
simple60_reason_05	CUDA OOM	No valid final response was obtained.	0.4 s
simple60_reason_06	CUDA OOM	No valid final response was obtained.	0.4 s
simple60_reason_07	CUDA OOM	No valid final response was obtained.	0.4 s

Appendix Original uniform results 41 to 60

Questions are identical to the two main conditions and can be located by ID in the preceding appendices. This table accounts for all 60 original results without interpreting OOM as a failure to understand the video. No row contains a valid semantic success.

ID	Outcome	Result content	Time
simple60_reason_08	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_reason_09	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_onset_01	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_onset_02	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_onset_03	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_onset_04	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_onset_05	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_onset_06	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_onset_07	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_onset_08	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_onset_09	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_negative_01	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_negative_02	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_negative_03	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_negative_04	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_negative_05	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_negative_06	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_negative_07	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_negative_08	CUDA OOM	No valid final response was obtained.	0.5 s
simple60_negative_09	CUDA OOM	No valid final response was obtained.	0.5 s